

COURSE CODE/TITLE:	CPE 010 - Data Structures and Algorithms	SECTION:	CPE21S1
DATE PERFORMED:	August 18, 2026	DATE SUBMITTED:	August 18, 2026
NAME/S:	Campo, Ramon Brendt D. Canoy, Hail B. Castillo, Kenneth (Absent) Corro, Yazinel De Leon, Benj Federic C.	INSTRUCTOR:	Engr. Jimlord M. Quejado

ACTIVITY NO. 6.1 Linear and Binary Search

1. PROCEDURE OUTPUT

Answer the following questions:

1. What is a search tree in data structures?

- A search tree is a form of tree data structure that stores data in a particular order to allow for more efficient searching, insertion and deletion of elements. One such type is Binary Search Tree. Each node in a BST has at most two children. Values less than the node are put into the left subtree, and values greater than the node are put into the right subtree.

2. What are the different types of search algorithms in data structures? Differentiate each type of search.

Sequential search

-it checks every element in a list/array from start to finish until it finds the element its looking for

Interval search

-divides the search space in half until it finds the position of the data its looking for

-(only applies to a sorted data structure)

tree/graph search

-Searches the data in a non linear data structure (trees and graphs)

There are 2 types of tree/graph search algorithms

1.breadth first search - uses a queue to search each node level by level

2.depth first search - uses a stack or recursion to search the data

3. What operations / implementations can be performed using binary and linear search operations?

Linear Search: The function iterates through the array from the first index to the last, comparing each element to the target value; if a match is found, it immediately returns that index, and if the loop finishes without finding a match, it returns -1.

Binary Search: The function starts with two pointers, left set to 0 and right set to the last index, and repeatedly calculates a middle index while left is less than or equal to right; it compares the middle element to the target, returning the middle index on a match, moving left to just past the middle if the target is larger, or moving right to just before the middle if the target is smaller, and returns -1 if the search range is exhausted without a match.

4. What are the advantages in using binary search tree as data structure?

A Binary Search Tree (BST) keeps data organized, it's useful and efficient to use. Using Binary Search Tree (BST) has a lot of advantages like fast searching, efficient insertion or deletion and many more.

1. **Fast Searching** - In a balanced BST it can find values easily because it removes half of the possible choices in each step.
2. **Easy Insertion** - New values can be added easily in which smaller values go left and the larger values go right.
3. **Easy Deletion** - A value can be removed in the tree while maintaining the tree organized.
4. **Keeps data in order** - Using inorder traversal allows the values to be displayed from smallest to largest to make it organized.
5. **Can grow or shrink** - Unlike arrays that have fixed size a BST can grow or shrink nodes when needed.
6. **Good for large data** - Even when big data as long as it's in order, it's organized and the tree is balanced and the BST can find values faster than checking every element 1 by 1

5. Give an example program using binary search and Linear search.

An example program demonstrating both algorithms in C++ features a linearSearch function that uses a for loop to check each array element sequentially from index 0 to n-1, alongside a binarySearch function that uses a while(left <= right) loop to repeatedly divide a sorted array search space in half by adjusting left and right index boundaries relative to a calculated mid position; the main function initializes a sorted vector ({12, 23, 34, 45, 56, 67, 78, 89, 90}), runs both algorithms to find a target value like 56, and prints the matching index location or returns -1 if the value is missing.

References:

11.4. Binary Search Trees — CS5020 Software Design & Data Structures. (n.d.).

<https://opendsa.server.cs.vt.edu/ODSA/Books/CS5020/html/BST.html?>

Sudhiksha.Malla, Search Algorithms: Sequential Search vs Interval Search,
<https://www.programming9.com/tutorials/competitive-programming/433-search-algorithms-sequential-search-vs-interval-search>

<https://www.youtube.com/watch?v=yYRMeEHbj6s>

Geeks4geeks, 2025, Difference Between BFS and DFS
<https://www.geeksforgeeks.org/cpp/cpp-program-for-bfs-traversal/>

University of Wisconsin-Madison. (n.d.) Binary searches trees. Department of Computer Sciences.

[Binary Search Trees](#)

GeeksforGeeks. (2024, May 14). *Searching algorithms*. geeksforgeeks.org
<https://www.geeksforgeeks.org/dsa/searching-algorithms/>

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____



COMPUTER
ENGINEERING

QUEZON CITY

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.